Quality Information Document

Ocean Monitoring Indicator (OMI) of the Copernicus Marine Environment Monitoring Service

Ocean Health Chl BALTIC OceanColour

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RECORD TABLE

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	2021/01/15	All	Creation of the document. The content has been extracted and	E. Böhm, V.Brando, S. Colella, M. Sammartino	
2.0	2021/12/15	All	Update for EIS December 2021	S. Colella	R. Santoleri
3.0	2022/06/16	All	EIS November 2022 – New template	Simone Colella	V. Brando
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5.0	2024/05/31	All	EIS November 2024	Simone Colella	V. Brando

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I INTRODUCTION

I.1 Product name

Product name	Dataset name
OMI_HEALTH_CHL_BALTIC_OceanColour_trend	omi_health_chl_baltic_oceanclour_trend
OMI_HEALTH_CHL_BALTIC_OceanColour_area_averaged_mean	omi_health_chl_baltic_oceanclour_area_averaged_mean
OMI_HEALTH_CHL_BALTIC_OceanColour_area_averaged_anomaly	N/A

I.2 Indicator evaluation method

The CMEMS Ocean Colour OMIs are derived from the reprocessed (MY) regional gridded chlorophyll products generated and distributed by the CMEMS OC-TAC. The input dataset are the spectral remote sensing reflectances (Rrs) produced by the Plymouth Marine Laboratory (PML) using an ad-hoc configuration for CMEMS of the ESA OC-CCI processor version 6 (OC-CCIv6) to merge at 1km resolution (rather than at 4km as for OC-CCI) MERIS, MODIS-AQUA, SeaWiFS, NPP-VIIRS and OLCI-A data. The chlorophyll product is derived from a Multi-Layer Perceptron neural-net (MLP) developed on field measurements collected within the BiOMaP program of JRC/EC (Zibordi et al., 2011). The algorithm is an ensemble of different MLPs that use Rrs at different wavelengths as input. The processing chain and the techniques used to develop the algorithm are detailed in Brando et al. (2021; 2024).

The MY regional chlorophyll dataset was used to compute anomaly map, with respect to reference climatology (1997-2014), daily and monthly regional mean time series and trend map.

Trend map was obtained by applying Colella et al. (2016) methodology. Monthly chlorophyll maps, derived from MY data, are used to compute the trend for the period 1997-2023. Trend analysis has been performed by using the X-11 seasonal adjustment procedure (Pezzulli et al. 2005). The X-11 procedure assumes that the CHL time series Y_t can be decomposed into its three components: the seasonal signal S_t, the inter-annual component T_t, and an irregular component I_t:

$$Y_t = T_t + S_t + I_t$$

The deseasonalized time series is obtained through an iterative process (Colella et al., 2016), by using successive filters, from simple moving-average, seasonal moving-average and Henderson filters (Vantrepotte and Mélin, 2009). At the end of this process, a seasonal signal is given, which is subtracted to the original times series in order to obtain a deseasonalized signal. Then, to estimate the magnitude of trend and its significance, the Mann-Kendall test (Mann, 1945; Kendall, 1975) and Sens's method (Sen, 1968) are subsequently applied on the deseasonalized

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time series. The final trend map is given in terms of relative changes (i.e., % yr^{-1}) by dividing, for each pixel, the trend slope by the corresponding climatological chlorophyll value.

Product: OMI_HEALTH_CHL_BALTIC_OceanColour_trend

This product includes the trend map obtained by using the methodology described in Colella et al. (2016). Here, trend magnitude and its significance are estimated by applying the Mann-Kendall test and the Sens's method on the deseasonalized monthly time series, as obtained from the X-11 technique, to estimate, trend magnitude and its significance. The trend is expressed in % $year^{-1}$ that represents the relative changes (i.e., percentage) corresponding to the dimensional trend [mg m^{-3} y^{-1}] with respect to the chlorophyll climatology (1997-2014). Only significant trends ($p < 0.05$) are included.

Product: OMI_HEALTH_CHL_BALTIC_OceanColour_area_averaged_mean

This product includes the monthly mean time series obtained by computing the average of 2D monthly mean (weighted by pixel area) over the region of interest. The deseasonalized time series is obtained by applying the X-11 seasonal adjustment methodology on the original time series as described in Colella et al. (2016), and then Mann-Kendall test (Mann, 1945; Kendall, 1975) and Sens's method (Sen, 1968) are subsequently applied to obtain the magnitude of trend.

Product: OMI_HEALTH_CHL_BALTIC_OceanColour_area_averaged_anomaly

The regional annual chlorophyll anomaly is computed by subtracting a reference climatology (1997-2014) from the annual chlorophyll mean, on a pixel-by-pixel basis and in log10 space.

I.3 CMEMS products used as input for the computations

Product name	Dataset Name	Documentation
OCEANCOLOUR_BAL_BGC_L3_MY_009_134	cmems_obs-oc_bal_bgc-plankton_my_l4-multi-1km_P1M	QUID: https://catalogue.marine.copernicus.eu/documents/QUID/CMEMOC-SQO-009-131to134.pdf PUM: https://catalogue.marine.copernicus.eu/documents/PUM/CMEMOC-PUM.pdf

II REFERENCES

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